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Chap-05
Linear Equations

$$x_1 + 2x_2 + x_3 = 0$$
$$x_1 - 2x_2 + 2x_3 = 4$$
$$2x_1 + 12x_2 - 2x_3 = 4$$

$O(n^3)$

$a_{12}\ a_{13}$
$a_{21}\ a_{23}$
$a_{32}\ a_{33}$

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix}$$

A $\quad$ x $\quad$ b

$$Ax = b \Rightarrow x = A^{-1}b = \begin{bmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 11 \\ -2.5 \\ -6 \end{bmatrix}$$

Lower Triangular Matrix:

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$$

upper triangular matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

3 var
↓
9

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Forward Substitution

$l_{11}x_1 = b_1 \Rightarrow x_1 = \frac{b_1}{l_{11}}$ (1 Div)

$l_{21}x_1 + l_{22}x_2 = b_2 \Rightarrow x_2 = \frac{b_2 - l_{21}x_1}{l_{22}}$ (1 Div + 1 mult, +1-sub) = 3

$l_{31}x_1 + l_{32}x_2 + l_{33}x_3 = b_3 \Rightarrow x_3 = \frac{b_3 - l_{31}x_1 - l_{32}x_2}{l_{33}}$ {2 div + 2 mult + 2 sub} = 5

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n rows/variable → n^2

$$\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Backward substitution

$u_{33} \cdot x_3 = b_3 \Rightarrow x_3 = \frac{b_3}{u_{33}} \rightarrow 1$

$u_{22} x_2 + u_{23} x_3 = b_2 \Rightarrow x_2 = \frac{b_2 - u_{22} x_3}{u_{22}} \rightarrow 3$

$u_{11} x_1 + u_{12} x_2 + u_{13} x_3 = b_1 \Rightarrow x_1 = \frac{b_1 - u_{12} x_2 - u_{13} x_3}{u_{11}} \rightarrow 5$

3

1 + 0n

Ire

jth row = $1 + (j-1) + (j-1)$

$= 1 + 2(j-1)$

$= 1 + 2j - 2$

$= 2j - 1$

total num of operations = $\sum_{j=1}^{n} (2j - 1)$

$= \sum_{j=1}^{n} 2j - \sum_{j=1}^{n} 1$

$= 2 \sum_{j=1}^{n} j - \sum_{j=1}^{n} 1$

$= 2 \times \frac{n(n+1)}{2} - n$

$= n(n+1) - n$

$= n^2 + n - n$

$= n^2$

Step 1: Augmented matrix

pivot element

$$\left[\begin{array}{ccc|c} \textcircled{1} & 2 & 1 & 0 \\ \textcircled{1} & -2 & 2 & 4 \\ \textcircled{2} & \textcircled{12} & -2 & 4 \end{array}\right]$$

a_{21}

$$= \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -4 & 1 & 4 \\ \textcircled{2} & 12 & -2 & 4 \end{array}\right]$$

a_{31}

$$= \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -4 & 1 & 4 \\ 0 & \textcircled{8} & -4 & 4 \end{array}\right]$$

a_{32}

$$= \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -4 & 1 & 4 \\ 0 & 0 & -2 & 12 \end{array}\right]$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 12 \end{bmatrix}$$

$$-2x_3 = 12$$

$$x_3 = -6$$

$$-4x_2 + x_3 = 4$$

$$\Rightarrow -4x_2 - 6 = 4$$

$$\Rightarrow x_2 = \frac{10}{-4}$$

$$= -2.5$$

$$R_i = R_i - m_{ik} R_k$$

$$m_{ik} = \frac{a_{ik}}{a_{kk}}$$

a_{21} : $i = 2,\ k = 1$

$$R_2 = R_2 - m_{21} R_1$$

$$m_{21} = \frac{a_{21}}{a_{11}} = \frac{1}{1}$$

$$R_2 = R_2 - R_1$$

a_{31} $i = 3,\ k = 1$

$$R_3 = R_3 - m_{31} R_1$$

$$m_{31} = \frac{a_{31}}{a_{11}} = \frac{2}{1} = 2$$

$$R_3 = R_3 - 2R_1$$

$a_{32} = 8$, $i = 3$; $k = 2$

$$R_3 = R_3 - m_{32} R_2$$

$$m_{32} = \frac{a_{32}}{a_{22}} = \frac{8}{-4} = -2$$

$$R_3 = R_3 + 2R_2$$

$$x_1 + 2x_2 + x_3 = 0$$

$$x_1 + 2(-2.5) - 6 = 0$$

$$\therefore x_1 = 11$$

$$\left[\begin{array}{ccc|c} 1 & \textcircled{2} & \textcircled{1} & 0 \\ 1 & -2 & \textcircled{2} & 4 \\ 2 & 12 & -2 & 4 \end{array}\right]$$

$R_2 = R_2 - m_{23} R_3$

$R_1 = R_1 - m_{13} R_3$

$R_1 = R_1 - m_{12} R_2$

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LU Decomposition

$Ax = b$

$LUx = b$

$Ux = y$

$Ly = b$

L — Lower Triangular Matrix

U — Upper Triangular matrix

Frobenius matrix:

Unit Lower Triangular Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix}$$

Ex: $x_1 + 2x_2 + x_3 = 0$

$x_1 - 2x_2 + 2x_3 = 4$

$2x_1 + 12x_2 - 2x_3 = 4$

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix}$$

$A \qquad x \qquad b$

Calculate u:

1) $A^{(1)} = A = \begin{bmatrix} 1 & 2 & 1 \\ (1) & -2 & 2 \\ (2) & 12 & -2 \end{bmatrix}$

$$F^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ -m_{21} & 1 & 0 \\ -m_{31} & 0 & 1 \end{bmatrix}$$

$$m_{21} = \frac{a_{21}}{a_{11}} = \frac{1}{1} = 1$$

$$m_{31} = \frac{a_{31}}{a_{11}} = \frac{2}{1} = 2$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

ii) $A^{(2)} = F^{(1)} A^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{bmatrix}$

$= \begin{bmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 8 & -4 \end{bmatrix}$

$F^{(2)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -m_{32} & 1 \end{bmatrix}$ $\quad m_{32} = \frac{a_{32}}{a_{22}} = \frac{8}{-4} = -2$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$

iii) $A^{(3)} = F^{(2)} \times A^{(2)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 8 & -4 \end{bmatrix}$

$U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 0 & -2 \end{bmatrix}$

$L = \begin{bmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -2 & 1 \end{bmatrix}$

$Ax = b$

$L(Ux) = b$

$Ux = b$

$Ly = b$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -2 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix}$$

$y_1 = 0$

$y_1 + y_2 = 4 \quad \therefore y_2 = 4$

$2y_1 - 2y_2 + y_3 = 4$

$\Rightarrow -8 + y_3 = 4 \quad \therefore y_3 = 12$

$$\therefore y = \begin{bmatrix} 0 \\ 4 \\ 12 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 12 \end{bmatrix}$$

$-2x_3 = 12 \quad \therefore x_3 = -6$

$-4x_2 + x_3 = 4 \quad \therefore x_2 = -2.5$

$x_1 + 2x_2 + x_3 = 0 \quad \therefore x_1 = 11$

$$x = \begin{bmatrix} 11 \\ -2.5 \\ -6 \end{bmatrix}$$

Proof:

$A^{(2)} = A^{(1)} \times A^{(1)}$

$A^{(3)} = F^{(2)} \times A^{(2)}$

$U = F^{(2)} \times F^{(1)} \times A^{(1)}$

$A^{(n)} = U = F^{(n-1)} \times F^{(n-2)} \times \dots \dots F^{(2)} F^{(1)} A^{(1)}$

$$F^{(n-1)} F^{(n-2)} \dots F^{(2)} F^{(1)} A^{(1)} = U$$

$$(ABC)^{-1} = C^{-1} B^{-1} A^{-1}$$

$$\Rightarrow A^{(1)} = \left(F^{(n-1)} F^{(n-2)} \dots F^{(2)} F^{(1)} \right)^{-1} U$$

$$\Rightarrow LU = F^{(1)^{-1}} F^{(2)^{-1}} \dots F^{(n-2)^{-1}} F^{(n-1)^{-1}} U$$

$$\Rightarrow \boxed{L = F^{(1)^{-1}} F^{(2)^{-1}} \dots F^{(n-2)^{-1}} F^{(n-1)^{-1}}}$$

Pivoting:

$$2x_2 + x_3 = 4$$

$$x_1 - 2x_2 + 2x_3 = 2$$

$$2x_1 + 12x_2 - 2x_3 = 4$$

$$\begin{bmatrix} 0 & 2 & 1 & 4 \\ 1 & -2 & 2 & 2 \\ 2 & 12 & -2 & 4 \end{bmatrix}$$

a_{21}

$$R_2 = R_2 - m_{21} R_1$$

$$m_{21} = \frac{a_{21}}{a_{11}}$$

Row swap

$$\begin{bmatrix} 1 & -2 & 2 & 2 \\ 0 & 2 & 1 & 4 \\ 2 & 12 & -2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 2 \\ 0 & 2 & 1 \\ 2 & 12 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix}$$

Column swap

$$\begin{bmatrix} 2 & 0 & 1 \\ -2 & 1 & 2 \\ 12 & 2 & -2 \end{bmatrix} \begin{bmatrix} x_2 \\ x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 4 \end{bmatrix}$$